HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

AI-Based Digital Twin Prototype for Two-Hour Glucose Forecasting

An experimental glucose forecasting proof of concept

TEAM	Transcendental-Coders
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COLLEGE	Kamla Nehru Institute of Technology
INCUBATOR	N/A



Experimental prototype - not clinically validated. Based on historical datasets.

The challenge: making sense of historical health data

Health records arrive as many files

EHR data comes as CSV files that must be inspected for missing values and duplicates.

Wearable glucose is a time series

Readings every 5 minutes need preprocessing and feature engineering before any forecasting.

The two datasets are not linked

Individual-level mapping between EHR and wearable data has not been established.

Broader Digital Twin vision (not claimed here)

A personalized model that combines health records and wearable data for an individual. This prototype does not establish patient-level integration, clinical validation, real-time monitoring or clinical decision support.

What this prototype actually implements

- EHR data quality pipeline
- Wearable glucose preprocessing and features
- Experimental two-hour glucose forecast
- Flask API serving readings and predictions
- Web dashboard for history and forecast

Project objectives

01

EHR data quality assessment

Inspect EHR CSV files and report missing values, duplicates and column-level quality.

02

Wearable glucose preprocessing

Inspect and prepare historical glucose time-series data.

03

Time-series feature engineering

Build lag, rolling, recent-change and activity features.

04

Experimental 2-hour forecasting

Predict glucose about 120 minutes ahead with a machine-learning model.

05

API-based prediction service

Serve participants, historical readings and forecasts through Flask.

06

Dashboard visualization

Let a user pick a participant and view history and forecast.

Two separate datasets

EHR dataset

18
EHR CSV files processed

0
processing failures (recorded run)

- Raw source data was preserved
- Checked for missing values, duplicates and column-level quality

processed separately

Wearable glucose dataset

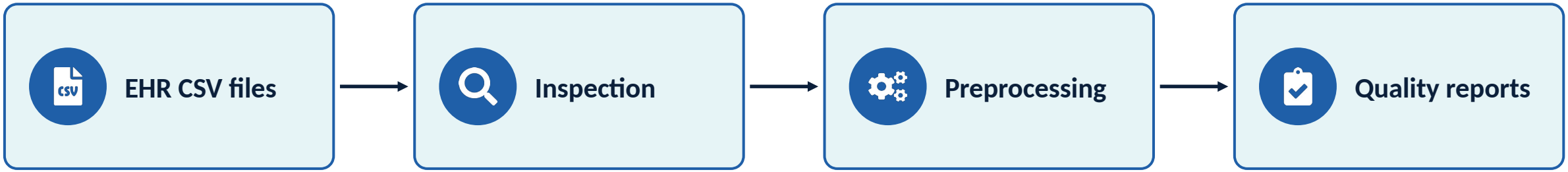
5 min
sampling interval

Steps, calories
activity features, where available

- Historical glucose time series
- Participants identified by IDs such as HUPA0001P

Individual-level mapping between the two datasets has not been established. No dataset statistics beyond those shown here are claimed.

EHR preprocessing pipeline



Missing-value analysis

Find gaps in each column of every EHR file.

Duplicate identification

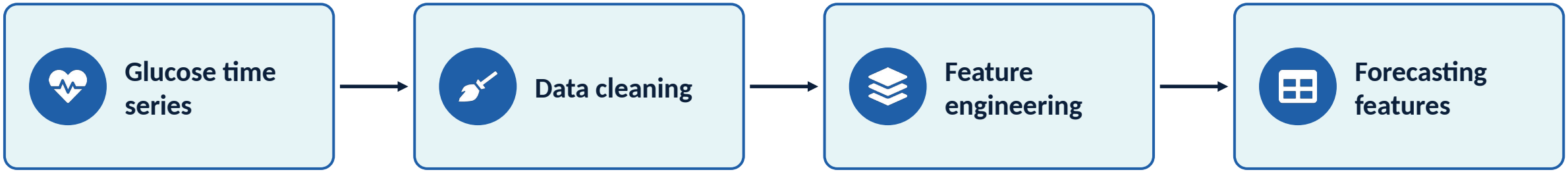
Identify duplicate records and remove them during preprocessing.

Column-level checks

Run data quality checks column by column and write the results to reports.

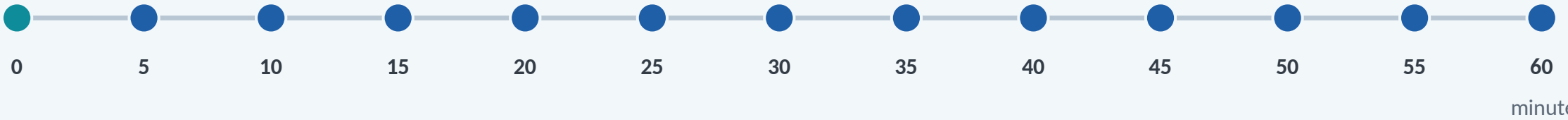
Output: cleaned EHR data and data quality reports. Raw source data is preserved.

Wearable data preprocessing



Sampling interval: one reading every 5 minutes

Each step is one reading; 12 readings cover one hour.



Preprocessing is applied to wearable glucose data only. EHR data is handled in a separate pipeline.

Feature engineering and the two-hour target

Historical lags

Earlier glucose readings give the model recent history.

Rolling statistics

Summary statistics over recent windows of readings.

Recent changes

How glucose has moved between recent readings.

Activity features

Steps and calories, where available.

Target construction

24 steps x 5 min = 120 min



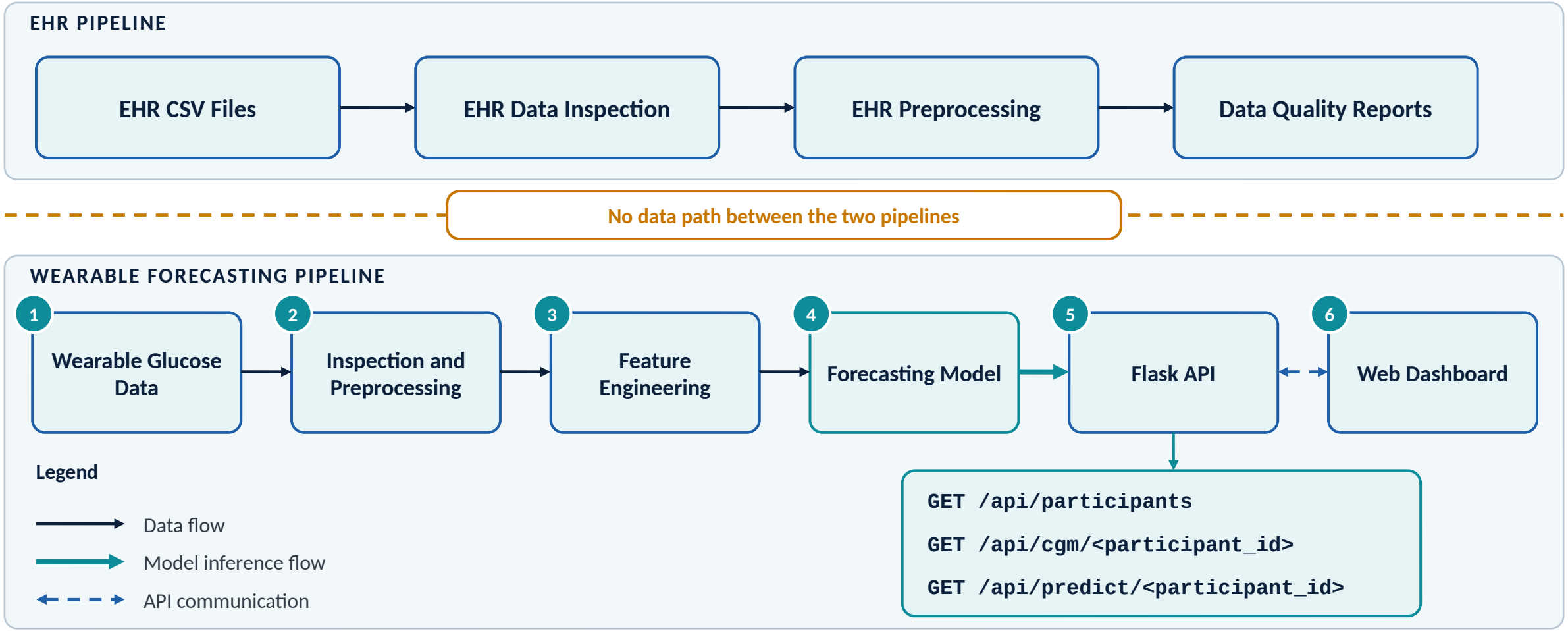
Now (t) **t + 24 steps**

Target = glucose about 120 minutes ahead (5-minute sampling)

Features: lags, rolling statistics, recent changes, activity

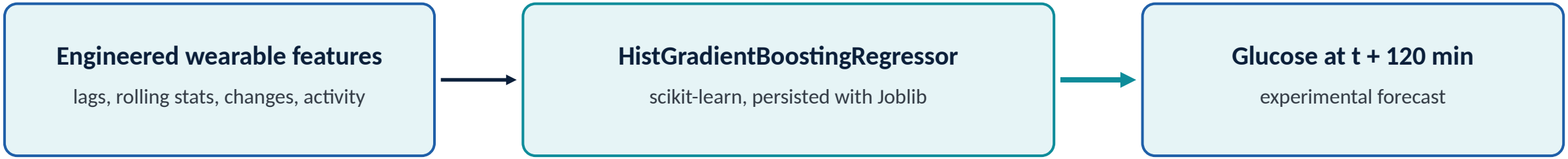
The future target is the prediction label only. It is never used as an input feature.

System architecture: two separate pipelines



EHR and wearable datasets are processed separately. Individual-level mapping has not been established.

Forecasting model and baseline



HistGradientBoostingRegressor

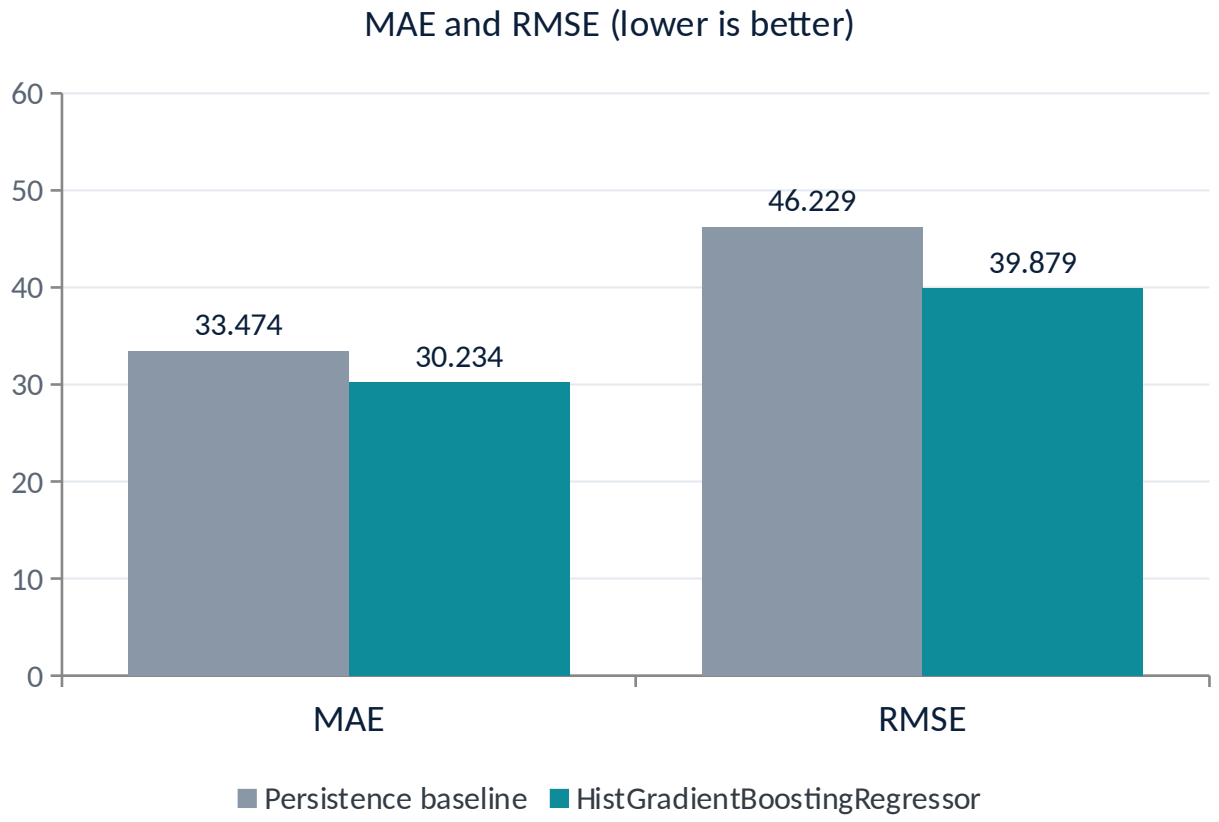
A scikit-learn model that builds many small decision trees one after another. Each new tree corrects the errors of the earlier ones. Here it learns from the engineered features to predict glucose 120 minutes ahead.

Persistence baseline

The simplest forecast: carry the current glucose value forward as the prediction. It gives a reference that the machine-learning model has to beat.

Experimental model. No clinical validation is claimed.

Recorded model evaluation



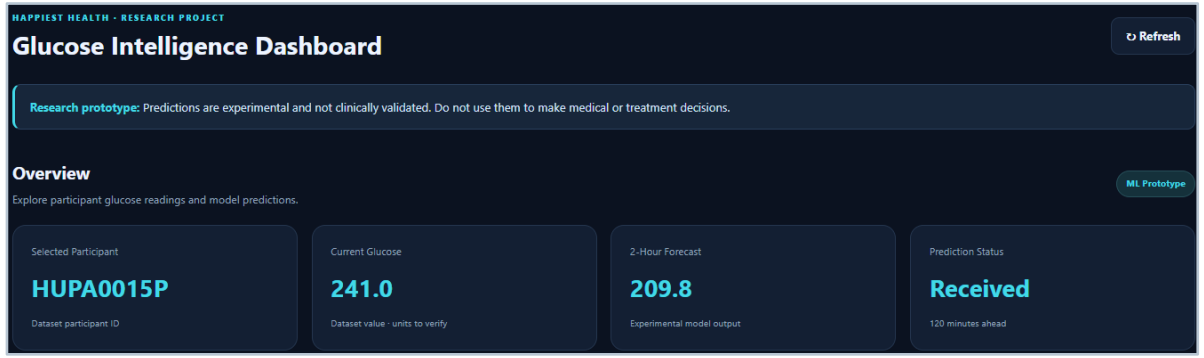
Metric	Persistence baseline	HistGradient-BoostingRegressor
MAE	33.474	**30.234**
RMSE	46.229	**39.879**
R^2	0.1376	**0.3583**

approx. 9.7%

lower MAE than the persistence baseline in the recorded evaluation

Recorded experimental evaluation results. Measurement units are not documented and are not stated. These results do not establish clinical usefulness or guaranteed generalization.

Working prototype: the web dashboard



Overview, participant HUPA0015P: current glucose 241.0, 2-hour forecast 209.8 (as displayed).



Participant selector and historical glucose chart (288 readings loaded).



ML Forecast panel: latest observed reading and the model's single two-hour forecast.

1. **Participant selection:** Pick a participant ID from the dropdown and press Load Participant.
2. **Historical glucose:** A line chart shows the observed readings for that participant.
3. **Forecast:** Shows the latest observed value and one two-hour forecast, with no intermediate predictions.

Flask API and a recorded prediction example

README.md | 127.0.0.1:5000 | git_diff_summary.txt U | LICENSE | .git

http://127.0.0.1:5000/

Pretty-print


```
{
  "history_endpoint": "/api/cgm/<participant_id>",
  "message": "Happiest Health Digital Twin API",
  "participants_endpoint": "/api/participants",
  "prediction_endpoint":
"/api/predict/<participant_id>",
  "status": "running"
}
```


Screenshot of the running API root page (127.0.0.1:5000): service name, status "running" and the endpoint listing.

Recorded example: HUPA0001P

Historical-dataset example, not a real-time clinical prediction

Current glucose	326.0
Predicted glucose after two hours	294.2235
Forecast horizon	120 minutes
Latest source timestamp	2018-06-27T23:55:00
Dataset status	historical_dataset
Model status	experimental_prototype

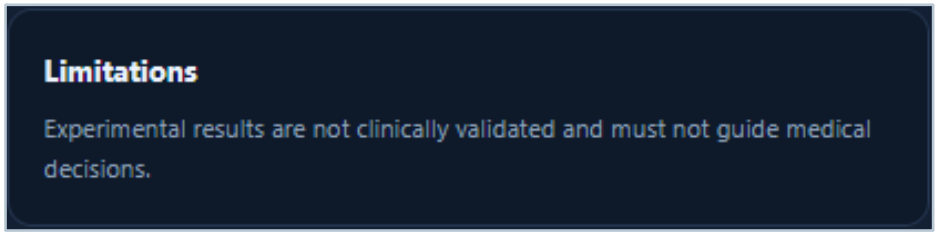
This HUPA0001P example comes from the architecture document and is separate from the HUPA0015P dashboard screenshots. It does not imply a medically meaningful result.

Limitations and future work

- **Not clinically validated.**
- Data is historical; this is not real-time monitoring.
- Individual-level mapping between EHR and wearable datasets has not been established.
- The current glucose model does not use EHR variables.
- Recorded performance does not establish clinical usefulness or guaranteed generalization.

Future work

- Further validation
- Data reconciliation, if valid patient-level linkage becomes available
- Additional participant-level evaluation



The dashboard shows the same caveat in its About This Project section.

Conclusion

EHR quality pipeline

Inspection, preprocessing and quality reports

Experimental forecast

Wearable features and HistGradientBoostingRegressor, 120 minutes ahead

Flask API

Three GET routes for participants, history and prediction

Web dashboard

Participant selection, historical readings and forecast view

A proof-of-concept workflow from data preparation to forecast visualization, built on historical datasets.

Repository: https://github.com/shubhampanchal6376/digital-twin

Team Transcendental-Coders | Team leader: Madhav Kumar | Kamla Nehru Institute of Technology

Disclaimer: experimental prototype, not clinically validated, historical data, no clinical decision support.